ML_CLASSIFICATION ALL ALGORITHMS DETAILED REPORT

BEFORE GRIDSEARCH_CV

SI_NO	NAME OF ALGORITHMS	LABLE PARAMETERS (USED FOR MODEL CREATION)	ACCURACY
1	Random Forest	n-estimator=10, criterion = 'entropy', random_state = 0	90%
2	Decision Tree	criterion = 'entropy', random_state = 0	87%
3	SVM	kernel = 'rbf', random_state = 0	78%
4	Logistic Regression	random_state=0	89%
5	KNN	n_neighbors=7,metric='minkowski', p=2	84%
6	MultinomialNB() naive_bayes	-----------	63
7	BernoulliNB	-----------	63
8	CategoricalNB	-----------	84
9	ComplementNB	-----------	51

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AFTER GRIDSEARCH_CV

SI_NO	NAME OF ALGORITHMS	LABLE PARAMETERS (USED FOR MODEL CREATION)	ACCURACY
1	Random Forest	{'criterion': 'entropy', 'max_depth': None, 'max_features': 'sqrt', 'n_estimators': 200}	93%
2	Decision Tree	{'criterion': 'gini', 'max_depth': 3, 'max_features': None}	90%
3	SVM	'C': 100, 'gamma': 'scale', 'kernel': 'rbf'	91%
4	Logistic Regression	'penalty': 'l2', 'solver': 'liblinear'	89%
5	KNN	{'metric': 'euclidean', 'n_neighbors': 5, 'p': 1, 'weights': 'uniform'}	92%
6	Multinomial Navi baye's	{'alpha': 0.1, 'fit_prior': True}	63%
7	BernoulliNB	print(grid.best_params_)	49%
8	CategoricalNB	{'alpha': 0.1, 'fit_prior': True}	83%
9	ComplementNB	{'alpha': 0.1, 'norm': False}	51%

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Best Performing Algorithm: Random Forest

Among the evaluated models, **Random Forest** algorithm is best algorithm because its accuracy **of 93%**. The model was fine-tuned using the following hyperparameters: criterion=**'entropy', max_depth=None, max_features='sqrt', and n_estimators=200**. I gained excellent experience working with all these classifier algorithms, especially when combined with StandardScaler. Initially, my model achieved only 90% accuracy, but after applying StandardScaler, the accuracy improved to 93%. It turned out to be a highly useful tool for me.